

PARAMETERIZATION, SURFACE AREA AND SURFACE INTEGRALS

1. Parameterize the **plane** passing through the origin with normal vector $\langle 1, 2, -1 \rangle$.
2. Find a parameterization for the **sphere** of radius a centered at the origin.
3. Parameterize the **cone** $z = \sqrt{x^2 + y^2}$ for $0 \leq z \leq 3$.
4. Find a parameterization for the **paraboloid** $z = x^2 + y^2$ over the disk $x^2 + y^2 \leq 4$.
5. The cylinder $x^2 + y^2 = 9$ extends from $z = 0$ to $z = 4$.
 - (a) Write its parameterization.
 - (b) Sketch or describe the surface orientation (outward normal).
6. Find the **surface area** of the part of the paraboloid $z = 9 - x^2 - y^2$ above the xy -plane.
7. Find the **surface area** of the cylinder $x^2 + y^2 = 4$ between $z = 0$ and $z = 3$.
8. Compute the surface area of the portion of the plane $z = 3x + 2y + 1$ above the region $0 \leq x \leq 1, 0 \leq y \leq 2$.
9. A sphere $x^2 + y^2 + z^2 = 9$ has its **upper hemisphere** cut by $z = 0$. Find its surface area.
10. Find the area of the **portion of the paraboloid** $z = x^2 + y^2$ bounded by $z = 1$.
11. Evaluate

$$\iint_S (x + y + z) dS$$

where S is the plane $z = 4 - x - y$ over the region $0 \leq x \leq 2, 0 \leq y \leq 1$.

12. Compute

$$\iint_S z^2 dS$$

for the part of the sphere $x^2 + y^2 + z^2 = 9$ above the xy -plane.

13. Find the surface integral

$$\iint_S (x^2 + y^2) dS$$

over the cylinder $x^2 + y^2 = 1, 0 \leq z \leq 2$.

14. Evaluate

$$\iint_S 1 dS$$

for the surface $z = 4 - x^2 - y^2$ above $z = 0$ (i.e., find total surface area of the paraboloid).

15. Compute the flux of $\vec{F} = (x, y, z)$ through the portion of the plane $x + y + z = 1$ in the first octant, oriented outward.
16. Find the flux of $\vec{F} = (0, 0, z)$ through the upper hemisphere $x^2 + y^2 + z^2 = 4$.
17. Evaluate the flux of $\vec{F} = (x, -y, 0)$ through the cylinder $x^2 + y^2 = 9$, $0 \leq z \leq 5$, oriented outward.
18. Find the outward flux of $\vec{F} = (yz, xz, xy)$ across the cube bounded by the planes $x = 0, y = 0, z = 0, x = 1, y = 1, z = 1$.
19. Compute the flux of $\vec{F} = (x, y, 2)$ through the surface $z = 1 - x - y$ over the region $0 \leq x \leq 1, 0 \leq y \leq 1$.
20. Let S be the part of the sphere $x^2 + y^2 + z^2 = 9$ in the first octant. Find the flux of $\vec{F} = (x, y, z)$ outward through S .